

Problem 2.45

KNOWN: A gas contained in a piston-cylinder assembly undergoes a constant-pressure expansion while being slowly heated. State data are provided.

FIND: For the gas, evaluate work and heat transfer. For the piston, evaluate work and change in potential energy.

SCHEMATIC & GIVEN DATA:

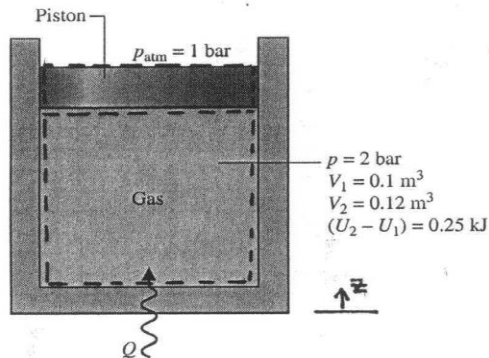


Fig. P2.56

ENGINEERING MODEL:

1. As shown in the schematic, two closed systems are considered: the gas and the piston.
2. The gas undergoes a constant-pressure process.
3. For the gas there is no change in potential energy (see Example 2.3) and no overall change in kinetic energy.
4. For the piston, there is no heat transfer. Also, there is no change in internal energy, no overall change in kinetic energy, and no friction.

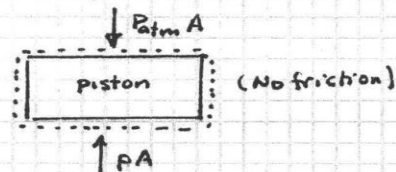
ANALYSIS: (a) Taking the gas as the system, the work is obtained from Eq. 2.17: $W = \int_1^2 p dV = p[V_2 - V_1] = (2 \times 10^5 \frac{N}{m^2})(0.12 - 0.1) m^3 \left| \frac{1 kJ}{10^3 N \cdot m} \right| = 4 kJ$

Reducing an energy balance, $\Delta U + \Delta KE + \Delta PE = Q - W \Rightarrow Q = W + \Delta U$

$$\Rightarrow Q = 4 kJ + 0.25 kJ = 4.25 kJ$$

(b) Taking the piston as the system, an energy transfer by work occurs on the bottom surface from the gas. At the top surface the piston does work on the atmosphere:

$$\begin{aligned} W_{piston} &= \int F dz = (P_{atm} A - P A) \Delta z = (P_{atm} - P) (\Delta z) \\ &= (P_{atm} - P) \Delta V_{gas} \\ &= (1 - 2) \left(10^5 \frac{N}{m^2} \right) (0.12 - 0.1) m^3 \left| \frac{1 kJ}{10^3 N \cdot m} \right| \\ &= -2 kJ \end{aligned}$$



An energy balance for the piston reduces as follows:

$$[\Delta U + \Delta KE + \Delta PE]_{piston} = Q_{piston} - W_{piston}$$

$$\Rightarrow \Delta PE]_{piston} = -W_{piston}$$

$$= + 2 kJ$$

①

1. Overall energy "balance sheet" in terms of magnitudes:

Input: $Q = 4.25 kJ$

Disposition of the energy input:

① Stored as ΔU in the gas:	0.25 kJ
② Stored as ΔPE in the piston:	2.00 kJ
③ Transfer by work to the atmosphere	2.00 kJ
	<u>4.25 kJ</u>

Problem 2.47

An electric motor operating at steady state draws a current of 20 amp at a voltage of 110 V. The output shaft rotates at a constant speed of 2000 RPM and exerts a torque of 9.07 N·m.

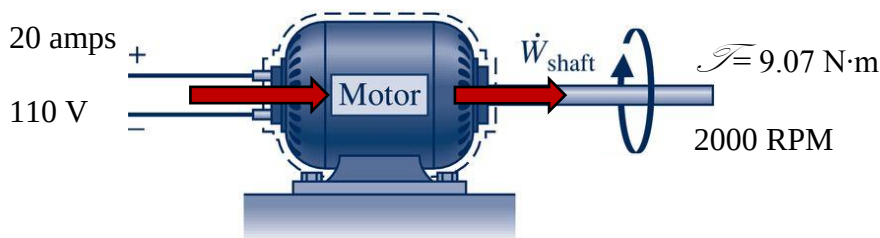
Determine

- (a) the magnitude of the power input, in W.
 - (b) the output power, in W.
 - (c) the cost of 24 hours of operation if electricity is valued at \$0.09 per kW·h.
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KNOWN: An electric motor operates at steady state. The input current and voltage are known and the rotational speed and torque of the exit shaft are specified. The cost of electricity is also specified.

FIND: Determine (a) the magnitude of the power input, (b) the output power, and (c) the cost of 24 hours of operation.

SCHEMCATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The motor operates at steady state.

ANALYSIS: (a) The magnitude of the input power is

$$|\dot{W}_{\text{elec}}| = (20 \text{ amp})(110 \text{ V}) \left| \frac{1 \text{ W}}{1 \text{ volt}\cdot\text{amp}} \right| \left| \frac{1 \text{ kW}}{10^3 \text{ W}} \right| = 2.2 \text{ kW (input)} \quad \leftarrow$$

(b) The shaft power is

$$\begin{aligned} |\dot{W}_{\text{shaft}}| &= \mathcal{T} \omega \\ &= (9.07 \text{ N}\cdot\text{m})(2000 \frac{\text{rev}}{\text{min}}) (2\pi \frac{1}{\text{rev}}) \left| \frac{1 \text{ min}}{60 \text{ s}} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N}\cdot\text{m}} \right| \left| \frac{1 \text{ kW}}{1 \text{ kJ/s}} \right| = 1.90 \text{ kW (out)} \quad \leftarrow \end{aligned}$$

(c) For steady operation (constant with time) the total amount of energy transferred in by electricity is

$$W_{\text{elec}} = \int |\dot{W}_{\text{elec}}| dt = |\dot{W}_{\text{elec}}| \Delta t = (2.2 \text{ kW})(24 \text{ h}) = 52.8 \text{ kW}\cdot\text{h} \quad \leftarrow$$

$$\text{Cost} = (52.8 \text{ kW}\cdot\text{h})(\$0.09/\text{kW}\cdot\text{h}) = \$4.75 \text{ (per day)}$$

COMMENT: Note, for steady state operation the inputs and outputs are constant with time.

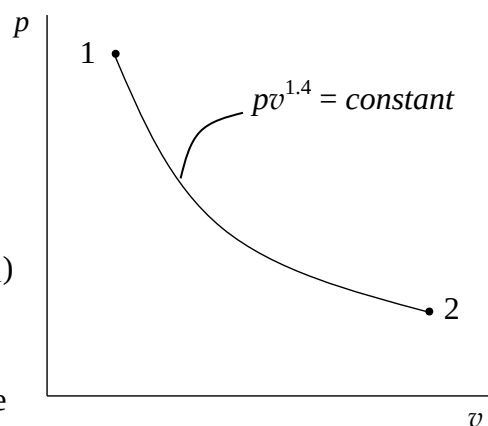
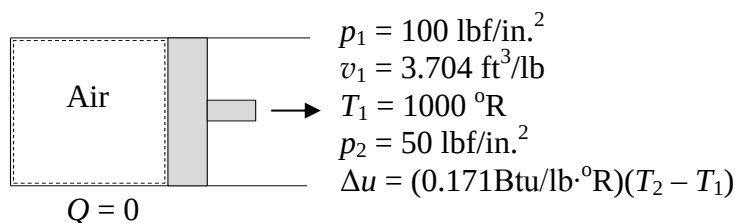
Problem 2.53

Air expands adiabatically in a piston-cylinder assembly from an initial state where $p_1 = 100$ lbf/in.², $v_1 = 3.704$ ft³/lb, and $T_1 = 1000$ °R, to a final state where $p_2 = 50$ lbf/in.². The process is polytropic with $n = 1.4$. The change in specific internal energy, in Btu/lb, can be expressed in terms of temperature change as $\Delta u = (0.171)(T_2 - T_1)$. Determine the final temperature, in °R. Kinetic and potential energy effects can be neglected.

KNOWN: Air undergoes a polytropic process with known n in a piston-cylinder assembly. Data are known at the initial and final states, and the change in specific internal energy is expressed as a function of temperature change.

FIND: Determine the final temperature.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The air is a closed system.

(2) The process is polytropic with $n = 1.4$ and volume change is the only work mode. (3) The process is adiabatic: $Q = 0$.

(4) Kinetic and potential energy effects are negligible.

ANALYSIS: To determine the final temperature, we will use the energy balance with the given expression for change in specific internal energy as a function of temperature change. First, determine the work using Eq. 2.17

$$W/m = \int_{v_1}^{v_2} p dv = \int_{v_1}^{v_2} \frac{\text{const}}{v^{1.4}} dv = \frac{(p_2 v_2 - p_1 v_1)}{1 - 1.4}$$

For the polytropic process, $p_1 v_1^{1.4} = p_2 v_2^{1.4}$. Thus

$$v_2 = \left(\frac{p_1}{p_2} \right)^{\frac{1}{1.4}} v_1 = \left(\frac{100 \text{ lbf/in.}^2}{50 \text{ lbf/in.}^2} \right)^{\frac{1}{1.4}} (3.704 \text{ ft}^3/\text{lb}) = 6.077 \text{ ft}^3/\text{lb}$$

So, the work is

$$W/m = \frac{(50 \text{ lbf/in.}^2)(6.077 \text{ ft}^3/\text{lb}) - (100)(3.704)}{1 - 1.4} \left| \frac{144 \text{ in.}^2}{1 \text{ ft}^2} \right| \left| \frac{1 \text{ Btu}}{778 \text{ ft} \cdot \text{lbf}} \right| = 30.794 \text{ Btu/lb}$$

The energy balance is: $\Delta U + \Delta KE + \Delta PE = Q - W$. With $\Delta U = m(u_2 - u_1)$

Problem 2.53 (Continued)

$$(u_2 - u_1) = -W/m$$

Inserting values

$$(0.171 \text{ Btu/lb} \cdot ^\circ\text{R})(T_2 - 1000 ^\circ\text{R}) = - (30.794 \text{ Btu/lb})$$

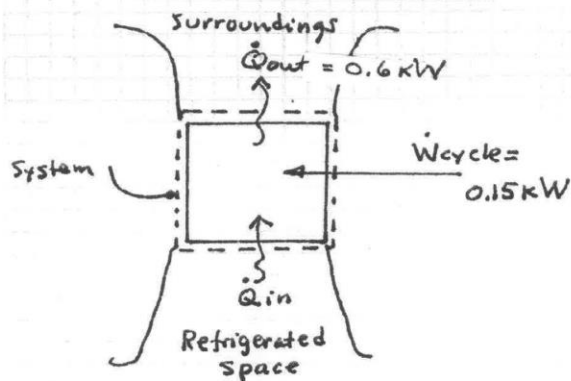
Solving; $T_2 = (-30.794 \text{ Btu/lb}) / (0.171 \text{ Btu/lb} \cdot ^\circ\text{R}) + 1000 = 819.9 ^\circ\text{R}$ ←

Problem 2.67

KNOWN: Steady-state operating data are provided for refrigerator.

FIND: Determine the rate energy is removed by heat transfer from the refrigerated space, in kW, and the coefficient of performance.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The system undergoes a refrigeration cycle.
2. Energy transfers are positive in the direction of the arrows on the schematic.
3. The cycle operates steadily.

ANALYSIS:

Applying Eq. 2.44 on a time rate basis:

$$\begin{aligned} \dot{W}_{\text{cycle}} &= \dot{Q}_{\text{out}} - \dot{Q}_{\text{in}} \\ \Rightarrow \dot{Q}_{\text{in}} &= \dot{Q}_{\text{out}} - \dot{W}_{\text{cycle}} \\ &= 0.6 \text{ kW} - 0.15 \text{ kW} \\ &= 0.45 \text{ kW} \end{aligned}$$

Then, with Eq. 2.45 on a time rate basis,

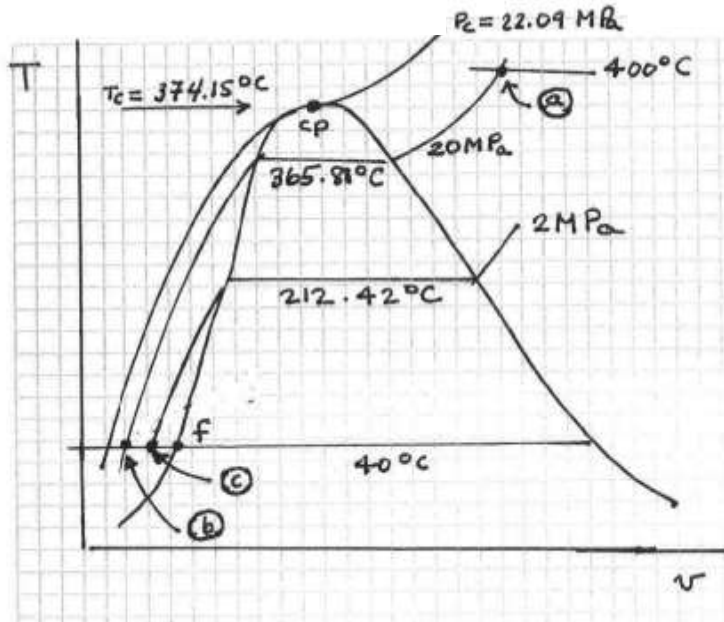
$$\begin{aligned} \beta &= \frac{\dot{Q}_{\text{in}}}{\dot{W}_{\text{cycle}}} \\ &= \frac{0.45 \text{ kW}}{0.15 \text{ kW}} \\ &= 3 \end{aligned}$$

← $\dot{Q}_{\text{in}}$

← β

PROBLEM 3.7 For H_2O , determine the specific volume at the indicated state, in m^3/kg . Locate the states on a sketch of the T - v diagram.

- (a) $T = 400^\circ\text{C}$, $p = 20 \text{ MPa}$.
- (b) $T = 40^\circ\text{C}$, $p = 20 \text{ MPa}$.
- (c) $T = 40^\circ\text{C}$, $p = 2 \text{ MPa}$.



(a) Table A-4

$$v = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$$

(b) Table A-5

$$v = 0.9992 \times 10^{-3} \text{ m}^3/\text{kg}$$

(c) Table A-2 using Eq. 3.11

$$v(T, p) \approx v_f(T) = 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$$